A Second Opinion for the Portfolio

How a New Forecasting Model Changes Cross-Asset Allocation Decisions

Sungkyu Lee | Graduate School of Computing, Yonsei University | Working Paper | October 2026

Investment Research · Empirical Research

Abstract

New forecasting models are often judged by prediction error. A portfolio, however, holds positions rather than forecasts. This paper examines that distinction in a monthly cross-asset ETF strategy, Forecast-to-Rank Allocation (F2R), which ranks 11 U.S.-listed ETFs and invests equally in the four highest-ranked assets. We compare its incumbent four-model ensemble with Chronos-2 and several matched combinations across 112 monthly decisions under common portfolio, execution, and cost rules.

Chronos-2 has a higher point-forecast RMSE (0.0570) than the conventional models (about 0.0550–0.0553), yet its standalone Top-4 portfolio reaches a 15.5% CAGR and 0.962 Sharpe ratio versus 14.9% and 0.891 for the incumbent. A hybrid combining three conventional models with Chronos-2 at a 20% rank-aggregation weight achieves a 17.1% CAGR and a Sharpe ratio of 0.987, with lower turnover than the incumbent. Relative to their respective conventional baselines, adding a 20% Chronos-2 rank contribution is associated with a 0.61-percentage-point CAGR improvement in the four-model ensemble and a 2.12-point improvement when Random Forest is excluded. With models and 80/20 combination weights unchanged, combining forecasts by rank rather than standardized magnitude is associated with a 2.53-percentage-point difference in CAGR, despite nearly identical turnover.

These are descriptive historical contrasts, not causal estimates. The exercise is a retrospective reconstruction using today's forecasting technology; Chronos-2 did not exist throughout 2017–2026, its pretraining corpus is not audited for temporal overlap, and the main hybrid was developed within the same historical research program rather than on an untouched holdout. The practical lesson is narrow: judge a new forecasting model by the portfolio decisions it changes, how its forecasts are integrated, and the consequences of those decisions—not only by average forecast error.

---

1. The Investment Question

An investor deciding whether to use a new forecasting model faces a different problem from a forecaster choosing the model with the smallest prediction error. Forecast accuracy matters, but it is only the first link in a longer chain. A new forecast must influence asset rankings enough to change holdings, and those holdings must deliver an attractive outcome after trading costs and risk are considered.

This distinction is familiar in decision theory and predict-then-optimize research: a statistically better estimate does not automatically produce a better downstream decision (Granger and Pesaran, 2000; Bertsimas and Kallus, 2020; Elmachtoub and Grigas, 2022). Asset-allocation research has also shown that machine-learning forecasts and forecast combinations can have economic value (Kynigakis and Panopoulou, 2022). This paper offers a more specific investment example: holding the portfolio rules fixed, it examines how the mix of forecasting models and the way their outputs are combined affect asset selection and portfolio results. The practical question is:

> **Does the new model improve the portfolio decision, or only the forecast?**

Takeaway. *A model that ranks poorly on forecast error may still deserve consideration if it improves the portfolio decisions that matter.*

2. The F2R Decision Process

F2R evaluates 11 U.S.-listed ETFs each month, ranks their expected returns, and assigns a 25% target weight to each of the four highest-ranked assets. In this study, the ETF universe, portfolio rule, trade timing, and transaction costs remain unchanged. Only the forecasting models and how their forecasts are combined differ. The universe spans U.S. large- and small-cap equities, international equities, real estate, U.S. Treasuries, broad commodities, and gold; Appendix A lists the ETF proxies.[1]

Combining models is not straightforward when their forecasts differ in scale and calibration. One approach first converts each model's predictions into rankings of the same 11 assets, then combines those rankings before selecting the portfolio. This gives the models a common basis for comparison, but it discards information about the size of predicted returns. Whether that trade-off helps the portfolio is a question for the evidence, not an assumption.

The four-asset cutoff makes even small ranking changes consequential. If the fourth- and fifth-ranked ETFs exchange places while the other three holdings are unchanged, 25% of the target portfolio shifts to a different asset. This is where a new forecasting model can supply a useful "second opinion."

3. What Exactly Changed in the Forecasting Ensemble?

The incumbent F2R ensemble combines four conventional machine-learning models. Two are regularized linear models (Ridge and Elastic Net), and two are tree-based nonlinear models (Random Forest and XGBoost). Their asset rankings are combined before the Top-4 portfolio is selected.

Chronos-2 is different because it is pretrained on time-series data rather than fitted only within the F2R research sample. It therefore brings information learned before being applied to this investment problem. In this experiment it is applied without additional task-specific training on the F2R sample (Ansari, Stella, et al., 2024; Ansari, Shchur, et al., 2025). The historical decision inputs are limited to prices available by each signal date, but this restriction does not establish point-in-time availability of the pretrained model. The conventional models use engineered price features fitted within F2R, while Chronos-2 uses daily log-return sequences and draws on pretraining information whose contents and possible overlap with the evaluation period are not audited here. Macroeconomic variables are excluded so that the comparison focuses on the forecasting approach rather than adding a new source of market information.

The six principal forecasting configurations are summarized below. Percentages refer to *weights assigned to forecasting models*, not to portfolio holdings.

Configuration	Forecasting components	Integration
Incumbent ensemble	Ridge, Elastic Net, Random Forest, XGBoost	Rank
Three-model ensemble	Ridge, Elastic Net, XGBoost	Rank
Chronos-2 alone	Chronos-2	Rank
Chronos + incumbent	Incumbent ensemble + 20% Chronos-2	Rank
Selected hybrid	Three-model ensemble + 20% Chronos-2	Rank
Standardized hybrid	Same as selected hybrid, same 80/20 weights	Standardized forecast magnitude

The selected hybrid is the main configuration examined in this paper; its 20% Chronos-2 weight should not be interpreted as an independently validated optimum.

[1] See Sungkyu Lee (2026), *The Decision Value of Price and Macro Information: Benchmark-Relative Evidence in Multi-Asset Allocation Using Historical-Vintage FRED-MD Data*, SlackQuant Investment Research Series, SSRN Abstract ID 7340100. That earlier study asks whether price or macro information adds decision value. The present paper keeps the portfolio-side information set price-only and instead studies model adoption and forecast integration.

We compare 112 monthly investment decisions from March 2017 through June 2026 under identical ETF, portfolio, execution, and cost rules. Both the incumbent ensemble and the selected hybrid reproduce their recorded Top-4 selections across all 112 decision dates. The comparison therefore holds portfolio construction and trading rules constant, allowing the differences between forecasting configurations to be examined on a consistent basis.

Evaluation and trading conventions. The 112 signal dates run from March 2017 through June 2026; portfolio metrics use 2,327 daily observations from May 1, 2017 through August 3, 2026. The first March 2017 signal executes in early April, but reported performance starts May 1 to exclude that initial partial month; the series extends through August 3 to complete the holding interval associated with the final June 2026 signal. Signals use the last common U.S. market close at or before month-end and execute at the first subsequent common close. Portfolio weights are allowed to drift with market movements between rebalances. One-way turnover is measured as half the total absolute change from pre-trade weights to target weights. Transaction costs are charged at 10 basis points on both purchases and sales. The Sharpe ratio is calculated from daily net returns, assuming a zero risk-free rate and annualizing by $\sqrt{252}$. Forecast error for every model is measured against the same next holding-period return target aligned to the execution clock. The model inputs are not identical: Chronos-2 uses the preceding 252 trading days of log returns and converts its median forecast path into a point signal for the subsequent holding period, while the conventional models use engineered price features. The rank-versus-standardized comparison fixes forecasts, models, 80/20 weights, Top-4 selection, execution, and costs and changes only the integration rule. Detailed ranking and tie-breaking rules, the underlying comparison data, and replication instructions accompany the research materials.

4. Forecast Accuracy Gives an Incomplete First Impression

Figure 1 compares each model's prediction error with the Sharpe ratio of the portfolio built from that model alone. The two measures give different rankings of the models.

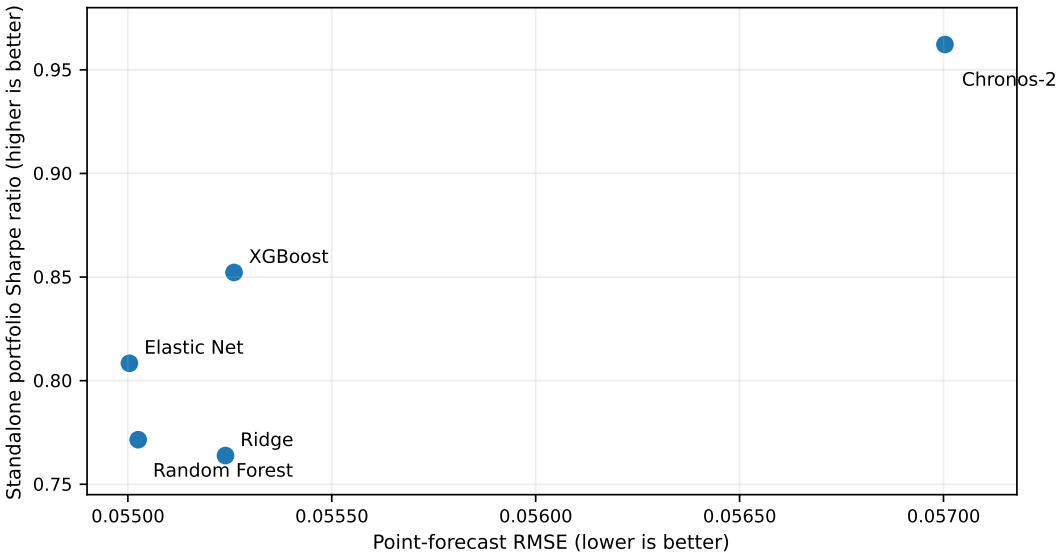


Figure 1. Forecast accuracy and standalone portfolio performance do not rank the models the same way. The horizontal axis reports root mean squared error (RMSE) across the common forecast panel, where lower is better. The vertical axis reports the Sharpe ratio, with the risk-free rate set to zero, of the Top-4 portfolio formed from each model's rankings under the same execution, cost, and accounting rules. Portfolio Sharpe ratios are computed from daily net returns, while model decisions are made at 112 monthly signal dates. The figure is descriptive; it does not imply that higher RMSE causes better portfolio performance.

Chronos-2 has the highest reported RMSE among the five standalone forecasting models at 0.0570. Elastic Net, Random Forest, Ridge, and XGBoost all cluster near 0.0550–0.0553. If the adoption decision stopped at point-forecast error, Chronos-2 would not be the obvious choice.

The portfolio ordering is different. Chronos-2 alone produces a 15.49% CAGR and a 0.962 Sharpe ratio; the

incumbent four-model ensemble produces 14.92% and 0.891 under the same accounting framework.

Prediction error remains relevant, but it is not a substitute for measuring the portfolio formed from those predictions. Decision-focused prediction emphasizes the same broad distinction: the forecasting loss and the downstream decision objective need not coincide (J. Lee et al., 2025). F2R does not train the models on a portfolio objective here; it shows the divergence in a controlled predict-then-rank setting after forecasts are translated into positions.

5. Where Did the Portfolio Improvement Come From?

Relative to the incumbent four-model ensemble, the selected hybrid has a 2.21-percentage-point higher historical CAGR and a 0.096 higher Sharpe ratio. Cumulative one-way turnover falls from 27.55 to 20.51 times portfolio value over the full period. This turnover measure adds up the value traded at each rebalance on a one-way basis; a figure of 1.0 corresponds to trading equal to one full portfolio value.

Because the selected hybrid both removes Random Forest and adds Chronos-2, the headline comparison alone cannot attribute the improvement. Figure 2 uses matched historical contrasts to separate those changes.

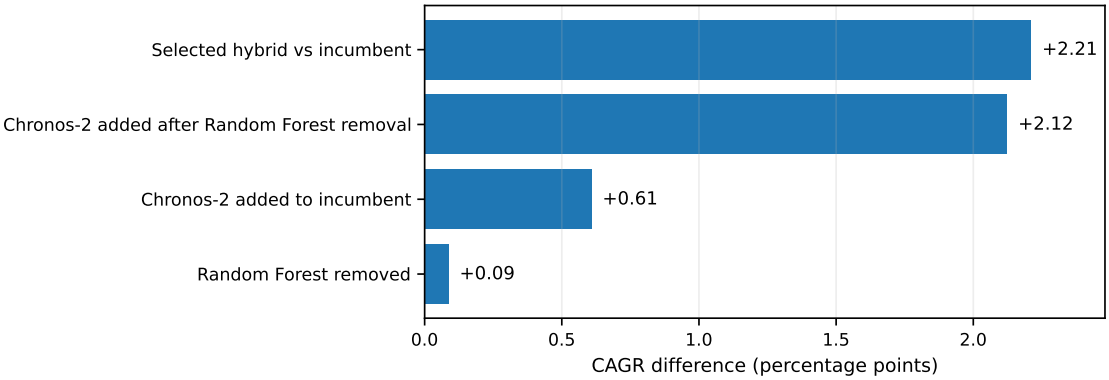


Figure 2. Sources of the historical portfolio improvement. Matched comparisons distinguish the contribution associated with removing Random Forest from that associated with adding a 20% Chronos-2 rank weight. Portfolio construction, execution, and cost assumptions are held constant. These differences are descriptive, not causal estimates.

Removing Random Forest alone is associated with a 0.09-percentage-point higher CAGR and a 0.022 lower Sharpe ratio. Adding a 20% Chronos-2 model-combination contribution to the unchanged four-model ensemble is associated with a 0.61-point higher CAGR and a 0.023 higher Sharpe ratio. After Random Forest removal, the same 20% contribution is associated with a 2.12-point higher CAGR and a 0.118 higher Sharpe ratio.

These results do not establish a general advantage for Chronos-2. Instead, they suggest that the investment value of an additional forecasting model depends on the models with which it is combined. In this historical comparison, the same 20% Chronos-2 contribution is associated with markedly different portfolio outcomes across the two conventional ensembles.

6. Where the Models Meet Matters

The selected hybrid combines relative asset rankings instead of averaging forecast magnitudes. The matched alternative uses the same three conventional models, the same Chronos-2 forecasts, and the same 80/20 model-combination weights, but instead standardizes each model's forecasts across the 11 ETFs before combining them. Standardization puts forecasts on a common scale while retaining information about relative magnitude.

Table 1. Same models and weights, different ways of combining forecasts.

	CAGR	Sharpe	Max DD	Turnover (× NAV)
Rank-based hybrid	17.13%	0.987	-21.97%	20.51
Standardized-forecast hybrid	14.60%	0.824	-24.55%	20.48

The rank-based variant has a 2.53-percentage-point higher CAGR and a 0.163 higher Sharpe ratio with nearly identical cumulative turnover. This is not universal proof that rank combination dominates. It shows that how forecasts are combined can materially affect the portfolio even when the underlying models are unchanged.

Rank aggregation can reduce differences in forecast scale and calibration, but it also discards information about forecast magnitude. The rank-based approach performs better in this historical comparison; whether that advantage generalizes remains uncertain.

7. The Second Opinion Matters When It Changes the Top-4

The selected hybrid does not create a different portfolio every month. Across 112 signal dates, its Top-4 set differs from the incumbent in 58 months (51.8%) and is identical in the remaining 54. Its average Top-4 Jaccard similarity with the incumbent is 0.783, where 1 means identical selections; similarity is higher against the three-model conventional ensemble, at 0.875. Most holdings therefore remain familiar; the difference arises when the new model pushes an asset into or out of the four-ETF portfolio.

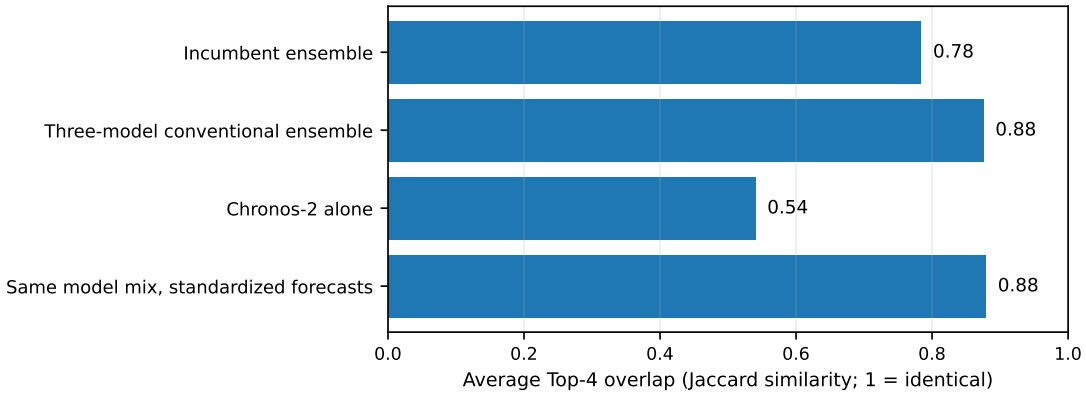


Figure 3. Most monthly Top-4 choices remain similar. Bars show average Jaccard overlap between the selected hybrid and each alternative across 112 monthly signal dates. A value of 1 means identical Top-4 sets. The figure measures decision similarity, not performance contribution.

Overlap does not show whether the altered holdings were profitable. To examine whether the changed selections helped, we attribute the hybrid's cumulative advantage over the incumbent to individual monthly portfolio decisions. Each contribution reflects differences in portfolio returns and associated trading costs, measured in log wealth. Together, these contributions reconcile to the observed wealth gap.

Those 58 changed-selection months account for 99.4% of the net relative log-wealth gap between the hybrid and the incumbent. Because both variants target the same equal-weight Top-4 portfolio when their selections coincide, the gross-return contribution in unchanged-selection months is zero by construction; the small residual comes only from path-dependent trading-cost differences. The 99.4% figure shows where the measured difference arises; by itself, it does not prove that the new model adds predictive value.

Of the 58 changed decisions, 36 contribute positively and 22 negatively. The largest favorable decision accounts for 10.7% of total positive changed-decision contribution. The five largest favorable decisions account for 35.2% of positive contribution but 71.3% of the net relative log-wealth gap, because negative changed decisions offset a substantial share of the gains. The result is not attributable to a single substitution, but the net advantage remains concentrated.

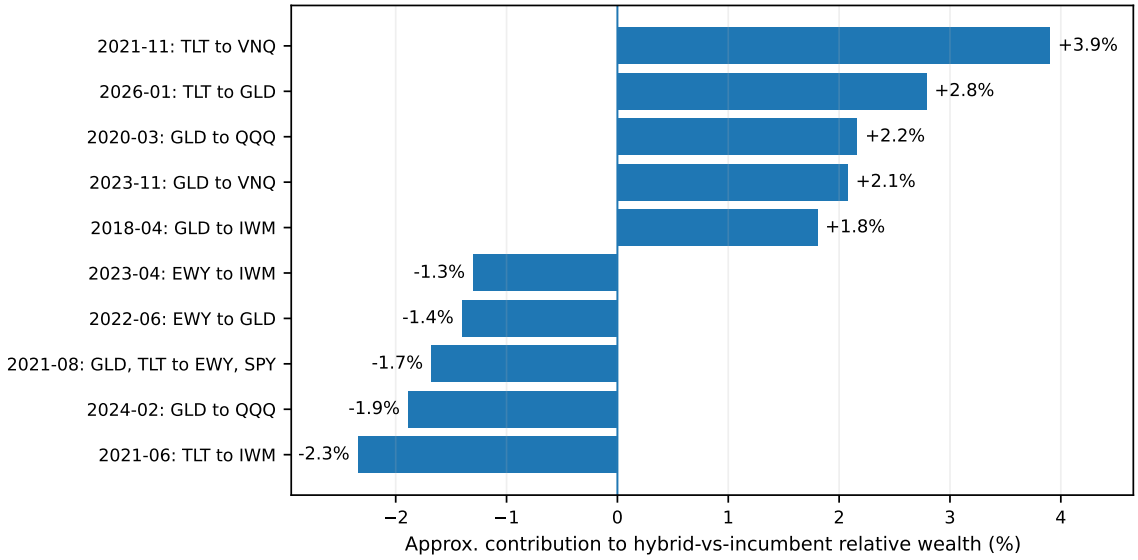


Figure 4. The changed decisions include both winners and losers. The figure shows the five largest favorable and five largest unfavorable changed Top-4 decisions. Each label gives the signal month and the incumbent asset or assets replaced by the hybrid. Bar labels convert each event's log-wealth contribution to an approximate relative-return percentage for readability; exact reconciliation is performed in log wealth. This is a post-hoc mechanism diagnostic, not independent out-of-sample evidence.

Figure 4 makes the trade-off tangible. Some ETF substitutions contributed materially to the hybrid's advantage, while others offset those gains. The value of a second opinion must therefore be judged by the investment decisions it changes, not by forecast accuracy alone. These retrospective contributions do not establish that the same decisions would succeed prospectively.

8. Higher Historical Returns Do Not Settle the Portfolio Choice

The selected hybrid does not lead every performance measure. A separate configuration combines the four conventional models in the incumbent ensemble with a 50% Chronos-2 model-combination weight. In this reconstruction, it has 17.63% CAGR and a 1.042 Sharpe ratio, above the selected hybrid's 17.13% and 0.987, but has a deeper maximum drawdown (-24.27% versus -21.97%). This four-conventional-model challenger differs from the three-conventional-model sensitivity tests in Figure 5.

Figure 5 reports that separate three-model comparison. Its 50% configuration has a 16.72% CAGR, not the 17.63% of the four-model candidate above; the existing 25% configuration has a 16.20% CAGR. Across the displayed three-model configurations, CAGR does not increase monotonically with the Chronos-2 contribution. These historical comparisons do not identify an optimal weight or validate a general model-adoption rule.

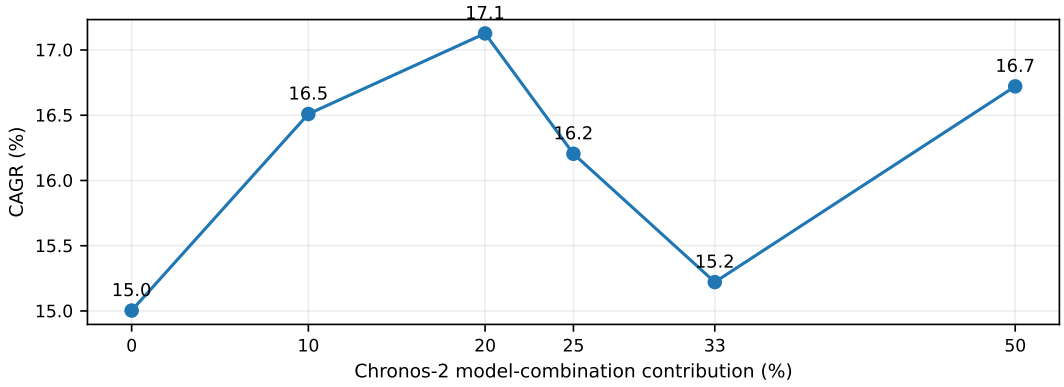


Figure 5. The historical result is not a smooth optimization curve. Within the same three-model conventional ensemble, the Chronos-2 model-combination contribution is varied across 0%, 10%, 20%, 25%, 33%, and 50%. CAGR is non-monotonic, so the selected 20% contribution should not be interpreted as an estimated optimum.

Figure 6 places the three relevant portfolio outcomes on a common cumulative-wealth scale. From an initial value of one, the incumbent reaches 3.62, the selected hybrid 4.32, and the four-model 50% Chronos-2 challenger 4.50. The selected hybrid's ending wealth is about 19.3% above the incumbent's. The challenger produces higher cumulative wealth and a higher Sharpe ratio, but also experiences a deeper maximum drawdown on the underlying *daily* return path. This trade-off illustrates why the highest historical return need not determine an investor's preferred portfolio. These paths use the frozen *monthly net-return observations*; they do not show within-month drawdowns or change the daily-frequency risk statistics used elsewhere in the paper.

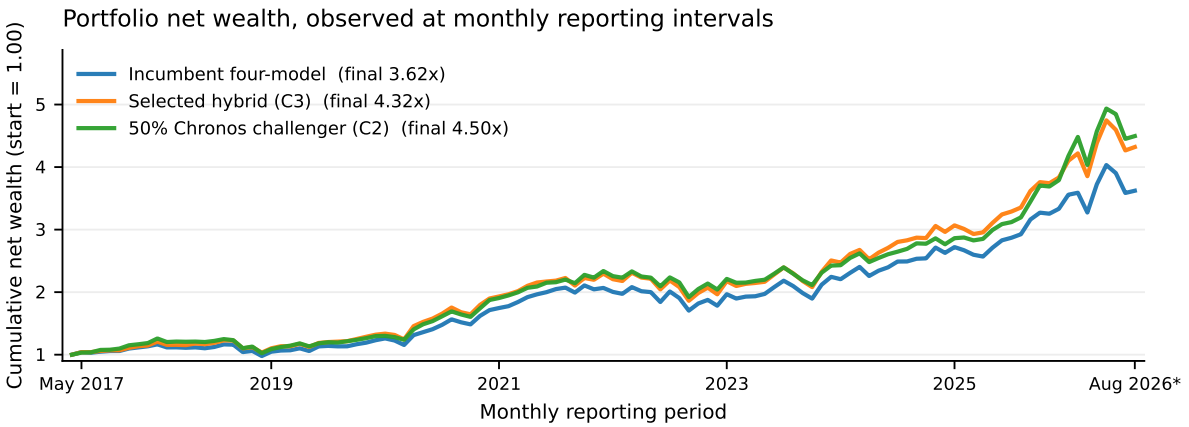


Figure 6. Cumulative net wealth across three portfolio configurations. The paths are constructed from the same frozen monthly net-return evidence, covering May 2017 through August 3, 2026. The final August observation is partial. The curves show monthly reporting values rather than daily performance or within-month drawdowns.

The selected 20% configuration should therefore be viewed as one historical balance of return, drawdown, and trading activity, not a proven optimum. The choice between it and the higher-return alternative depends on the investor's risk priorities as well as evidence beyond this retrospective sample.

> ***Takeaway.*** *Adding a forecasting model is a portfolio decision. Judge its value through the holdings it changes, their realized returns, trading costs, and the risks investors must bear—not simply the highest historical CAGR.*

9. What the Evidence Does Not Prove

The most important limitation is temporal. Chronos-2 did not exist throughout the 2017–2026 historical window, so this exercise does not claim that an investor could literally have run it in 2017. It is a retrospective comparison that applies today's model to past portfolio decisions, using the historical price information available at each signal date and the same trading rules. Because Chronos-2 is pretrained, however, this does not imply identical training information or prior knowledge across models; the study does not audit the contents or temporal overlap of its pretraining corpus. Accordingly, the Chronos-2 results should not be read as a real-time historical out-of-sample track record.

The selected hybrid emerged from exploratory work using the same historical evidence examined in this paper. Although matched comparisons help distinguish the contributions of individual design choices, they cannot eliminate selection bias or substitute for an independent evaluation period. Third, the evidence comes from one 11-asset universe, one Top-4 decision rule, and one pretrained model family. Fourth, the rank-versus-standardized-forecast comparison is informative inside this system but is not a universal test of forecast representation.

The decision-level attribution explains where the observed advantage arose, but it does not establish that similar substitutions will succeed in the future. The five largest favorable decisions account for 71.3% of the net relative log-wealth gap, highlighting the concentration of the historical result. Prospective results should be tracked separately from this reconstruction. Other universes and portfolio rules would provide more direct evidence of how broadly the findings apply. The present study is limited to price-based information; it does not answer the separate question of whether macroeconomic inputs improve these choices.

---

10. Practical Model-Adoption Checklist

The findings suggest a practical framework for evaluating any additional forecasting model in an investment process:

1. **Start with the portfolio decision.** What position, ranking, or allocation is the new model supposed to improve?
2. **Hold the investment setup fixed where possible.** Keep the investable universe, portfolio rules, execution timing, and transaction costs consistent across comparisons, while disclosing differences in model inputs and pretraining.
3. **Report forecast and portfolio metrics side by side.** Lower forecast error does not necessarily translate into better portfolio outcomes.
4. **Separate the sources of improvement.** Compare model additions, removals, and alternative ways of combining forecasts while holding the rest of the portfolio process fixed.
5. **Check what changed and where the gains came from.** In a strategy that holds only the highest-ranked assets, a small rank change near the cutoff can move a large portfolio weight, and a few decisions can dominate the net result.
6. **Do not turn one historical result into a universal rule.** A historical trade-off can motivate further use or testing without establishing an optimal configuration that will generalize.

11. Conclusion

The central finding is that forecast accuracy and portfolio value need not move together. Chronos-2 records a higher forecast error, yet the portfolio formed from its rankings achieves a higher historical Sharpe ratio than the incumbent ensemble. The same 20% Chronos-2 contribution is associated with a much larger historical improvement after the conventional ensemble is reconfigured than when it is appended to the incumbent ensemble. A second matched comparison shows that, with the same forecasts and model-combination weights, rank-based and standardized-forecast integration can produce materially different portfolio outcomes with almost identical turnover.

Model adoption cannot be settled by a ranking of forecasting errors alone. In the retrospective attribution, the selected hybrid differs from the incumbent in 58 of the 112 monthly decisions. These changed selections account for 99.4% of the observed net relative log-wealth gap. The five largest favorable decisions alone contribute 71.3% of that net gap, although unfavorable decisions offset part of the gains. These figures describe the historical sources of performance rather than establish a repeatable advantage. The investment question is whether a new model changes holdings in a useful way and whether those decisions remain attractive after costs, risk, and adverse periods are considered.

For investors and investment researchers, the durable principle is simple: **do not ask only whether a new model forecasts better. Ask what portfolio decision it changes, why it changes it, and whether that changed decision earns its keep.**

A. Investment Universe

Table 2 lists the 11 ETF proxies used in the model-adoption comparison. The broader investment context is discussed in the related allocation study cited in Section 2.

Table 2. ETFs included in the portfolio comparison.

Ticker	Economic exposure
SPY	U.S. large-cap equities (S&P 500)
QQQ	Nasdaq-100 equities
IWM	U.S. small-cap equities (Russell 2000)
EFA	Developed-market equities outside the U.S. and Canada
EEM	Emerging-market equities
EWY	South Korean equities
VNQ	U.S. listed real estate / REITs
IEF	Intermediate U.S. Treasuries (7–10 year)
TLT	Long-duration U.S. Treasuries (20+ year)
DBC	Broad commodities
GLD	Gold

References

Ansari, Abdul Fatir, Oleksandr Shchur, Jaris Küken, Andreas Auer, Boran Han, Pedro Mercado, Syama Sundar Rangapuram, Huibin Shen, Lorenzo Stella, Xiyuan Zhang, Mononito Goswami, Shubham Kapoor, Danielle C. Maddix, Pablo Guerron, Tony Hu, Junming Yin, Nick Erickson, Prateek Mutalik Desai, Hao Wang, Huzefa Rangwala, George Karypis, Yuyang Wang, and Michael Bohlke-Schneider (2025). "Chronos-2: From Univariate to Universal Forecasting". In: *arXiv preprint arXiv:2510.15821*.

Ansari, Abdul Fatir, Lorenzo Stella, Caner Turkmen, Xiyuan Zhang, Pedro Mercado, Huibin Shen, Oleksandr Shchur, Syama Sundar Rangapuram, Sebastian Pineda Arango, Shubham Kapoor, Jasper Zschiegner, Danielle C. Maddix, Hao Wang, Michael W. Mahoney, Kari Torkkola, Andrew Gordon Wilson, Michael Bohlke-Schneider, and Yuyang Wang (2024). "Chronos: Learning the Language of Time Series". In: *arXiv preprint arXiv:2403.07815*.

Bertsimas, Dimitris and Nathan Kallus (2020). "From Predictive to Prescriptive Analytics". In: *Management Science* 66.3, pp. 1025–1044. DOI: 10.1287/mnsc.2018.3253.

Elmachtoub, Adam N. and Paul Grigas (2022). "Smart 'Predict, then Optimize'". In: *Management Science* 68.1, pp. 9–26. DOI: 10.1287/mnsc.2020.3922.

Granger, Clive W. J. and M. Hashem Pesaran (2000). "Economic and Statistical Measures of Forecast Accuracy". In: *Journal of Forecasting* 19.7, pp. 537–560. DOI: 10.1002/1099-131X(200012)19:7<537::AID-FOR769>3.0.CO;2-G.

Kynigakis, Iason and Ekaterini Panopoulou (2022). "Does Model Complexity Add Value to Asset Allocation? Evidence from Machine Learning Forecasting Models". In: *Journal of Applied Econometrics* 37.3, pp. 603–639. DOI: 10.1002/jae.2885.

Lee, Junhyeong, Haeun Jeon, Hyunglip Bae, and Yongjae Lee (2025). "Return Prediction for Mean-Variance Portfolio Selection: How Decision-Focused Learning Shapes Forecasting Models". In: *Proceedings of the 6th ACM International Conference on AI in Finance*, pp. 114–122. DOI: 10.1145/3768292.3770423.

Lee, Sungkyu (2026). "The Decision Value of Price and Macro Information: Benchmark-Relative Evidence in Multi-Asset Allocation Using Historical-Vintage FRED-MD Data". SlackQuant Investment Research Series. SSRN Abstract ID 7340100.

Research materials and disclosure

Data and code availability. Public replication materials are available at github.com/slackquant80/f2r-model-adoption-replication. The accompanying release includes the frozen derived monthly net-return evidence and code to reproduce Figures 1–6, including the monthly-sampled cumulative-wealth comparison in Figure 6. These materials support the reported analysis but do not reproduce the original model training or the full historical portfolio simulation from raw market data. Versioned archives and DOI links are provided in the repository.

Research platform. Related research, replication resources, and interactive materials are available at research.slackquant.com.

Generative AI disclosure. OpenAI's ChatGPT assisted with coding and language editing.

The author takes full responsibility for this paper.